

Climate 5000 Series - Gen 4

9K BTU Single Zone Max Performance System

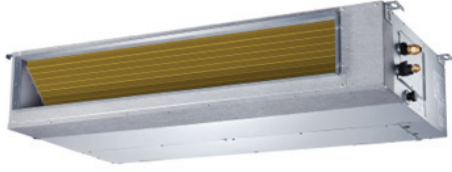
Ducted IDU

Engineering
 Submittal
 Sheet



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Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

9K Btu Single Zone Max Performance System - Ducted IDU

Indoor part number		8-733-962-709
Indoor model number		BMS500-AAU009-1AHDXD
Outdoor part number		8-733-962-688
Outdoor model number		BMS500-AAS009-1CSXHD
Efficiency Data		AHRI 210/240 2023
SEER2		21.4
EER2		13.4
HSPF2 Region IV		12.5
HSPF2 Region V		9.7
COP(47F)		3.53
COP (5F)		1.89
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3400~13500
	Rated Capacity	9000
	Current	4.8
	Input Power	667
Heating	Capacity Range	3800~16200
	Rated Capacity (47F)	10000
	Rated Capacity (5F)	11200
	Current	3.9
	Input Power	830
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 15.0
Max. fuse indoor/outdoor (A)		15.0 / 15.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 350
Input (W)		885
Rated current (RLA/A)		6.0
Piping		
Refrigerant precharge (oz/kg)		35.27 / 1.00
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		1/4" / Ø6.35
Gas side (in/mm)		3/8" / Ø9.52
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.16 / 15
Max. refrigerant pipe length (ft/m)		82.00 / 25
Max. difference in height (ft/m)		32.80 / 10

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Operation Range

ODU	Cooling (°F / °C)	-22 ~ 122 / -30 ~ 50
	Heating (°F / °C)	-22 ~ 75 / -30 ~ 24
IDU	Cooling (°F / °C)	60 ~ 90 / 16 ~ 32
	Heating (°F / °C)	32 ~ 86 / 0 ~ 30

System Data

Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		N/A
IDU	Fan motor model	ZKFN-81-8-1
	Fan motor qty (ea)	1
	Fan motor input (W)	150.0
	Fan speed, H / M / L (RPM)	980 / 910 / 840
	Air flow, Turbo / H / M / L / Si (CFM)	323.7 / 264.9 / 206.0 / 147.2 / 117.7
	Noise level, Turbo / H / M / L / Si (dB(A))	N/A / 32.5 / 31 / 29.5 / 23
	Net weight (lbs / kg)	39.46 / 17.9
	Gross weight (lbs / kg)	46.74 / 21.2
ODU	ESP range (Pa)	0-100
	Drain pipe outside diameter in. (mm)	OD 0.98 (25)
	Fan motor model	ZKFN-34-10-1-3
	Fan motor qty (ea)	1
	Fan motor input (W)	50
	Fan motor speed, H / / L (RPM)	850 / 680 / 450
	Air flow, Max (CFM)	1471.5
	Noise level, Max (dB(A))	56.0
Accessories	Net weight (lbs / kg)	72.75 / 33
	Gross weight (lbs / kg)	78.7 / 35.7

Accessories

BMS-WT2-XXD (8733962707)	Wired Wall Thermostat
BMS-DRY-XXD (8733969263)	24V Interface
BMS-PK1-7X1 (8733951010)	25ft Lineset

Bosch Thermotechnology Corp.
 Londonderry, NH • Watertown, MA • Ft. Lauderdale, FL

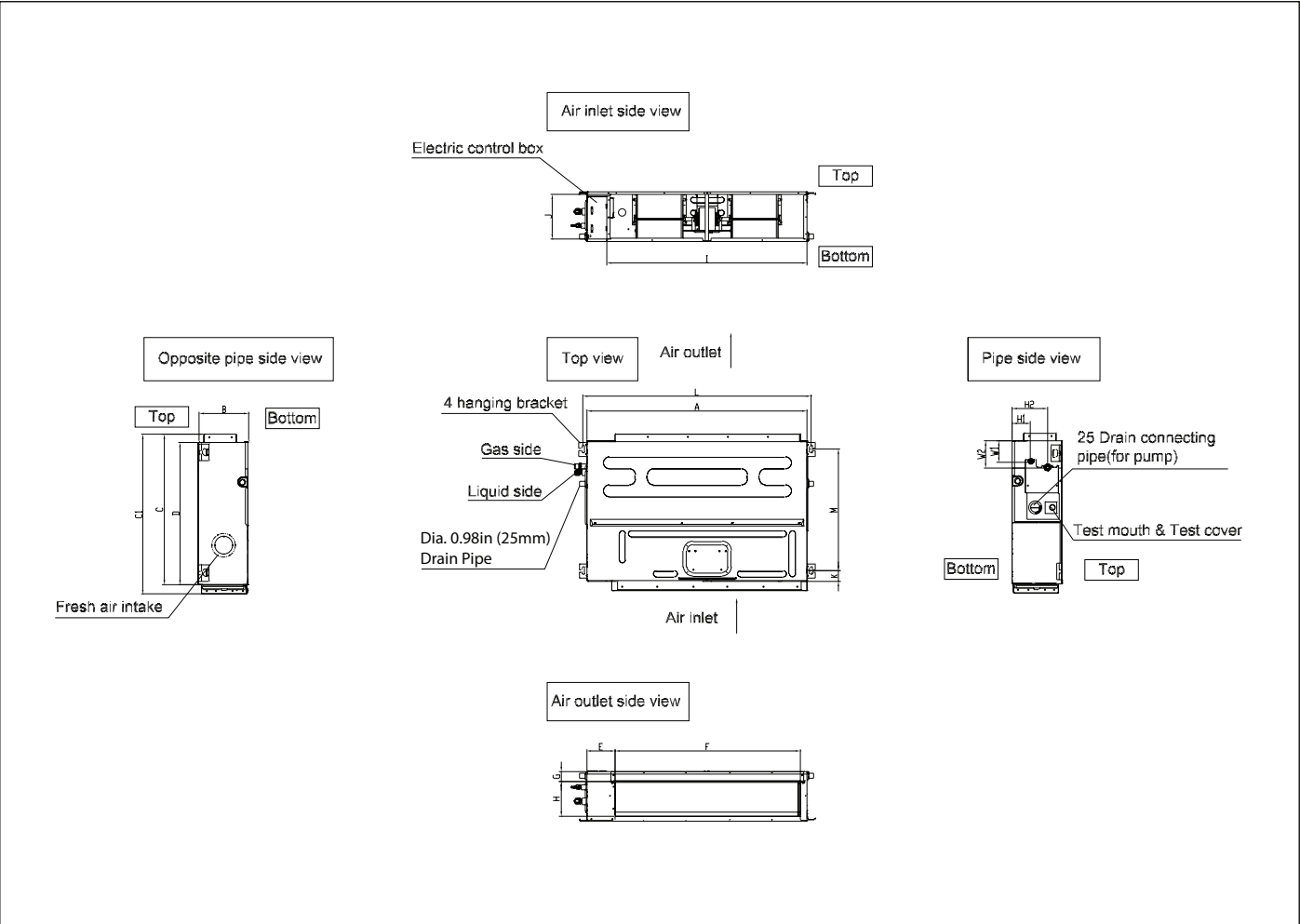
Bosch Thermotechnology Corp. reserves the right to make changes without notice due to continuing engineering and technological advances | BTC 769208348 B | 01.2026

Tel: 1-800-283-3787 www.bosch-homecomfort.us

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9K Btu Single Zone Max Performance System - Ducted IDU



Indoor Unit Dimensions and Connections

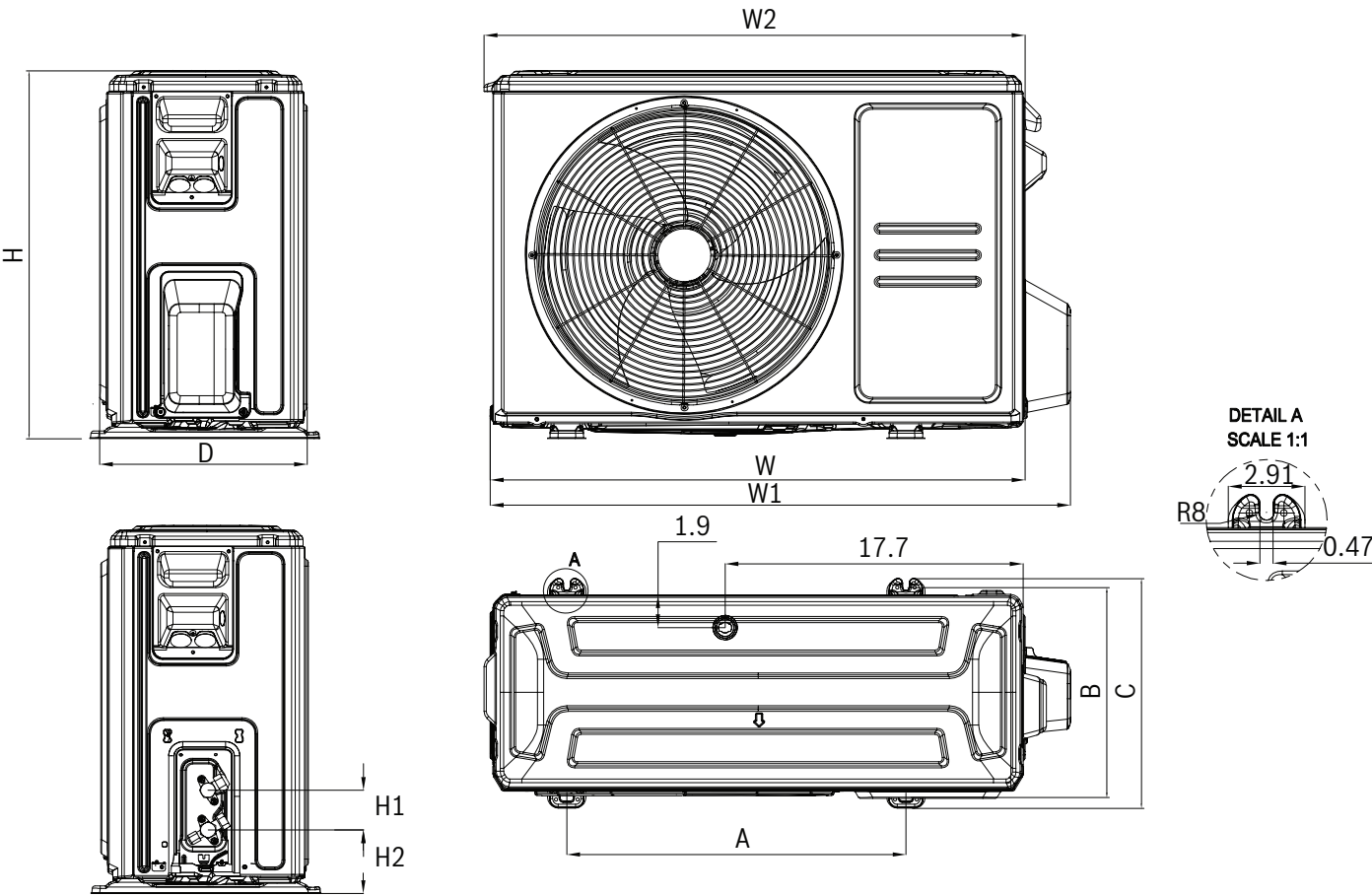


Model	Outline dimension (in)				Air outlet dimension (in)				Return air (in)			Mounting (in)		Gas and liquid valve (in)			
	A	B	C	D	E	F	G	H	I	J	K	L	M	H1	H2	W1	W2
BMS500-AAU009-1AHDXD	27.6	7.9	19.9	17.7	5.4	21.1	1.2	6.0	23.6	7.3	2.0	29.2	14.2	3.3	5.5	3.3	3.3

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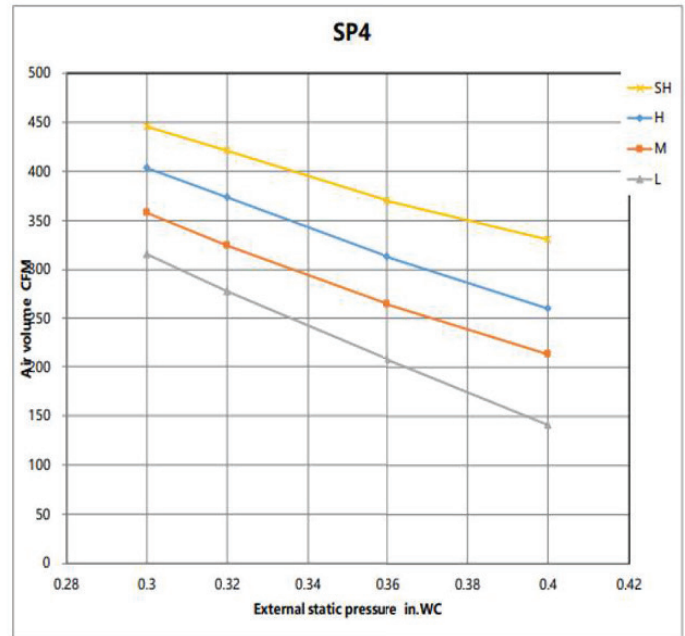
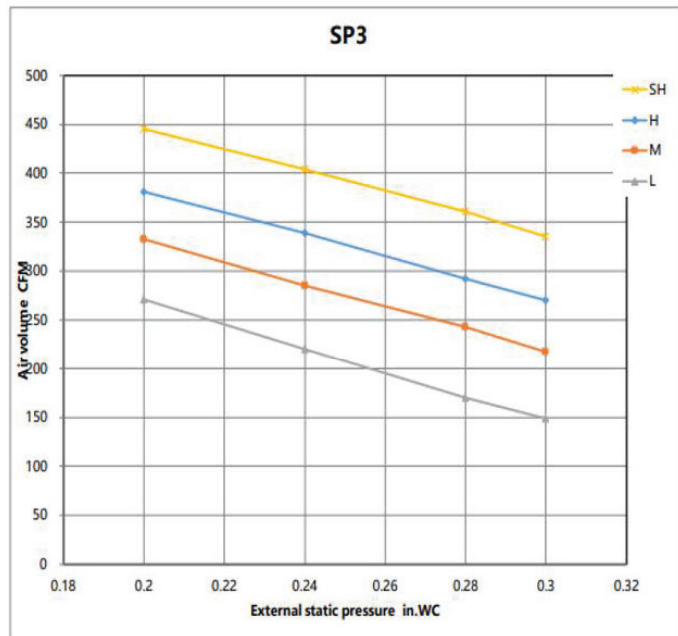
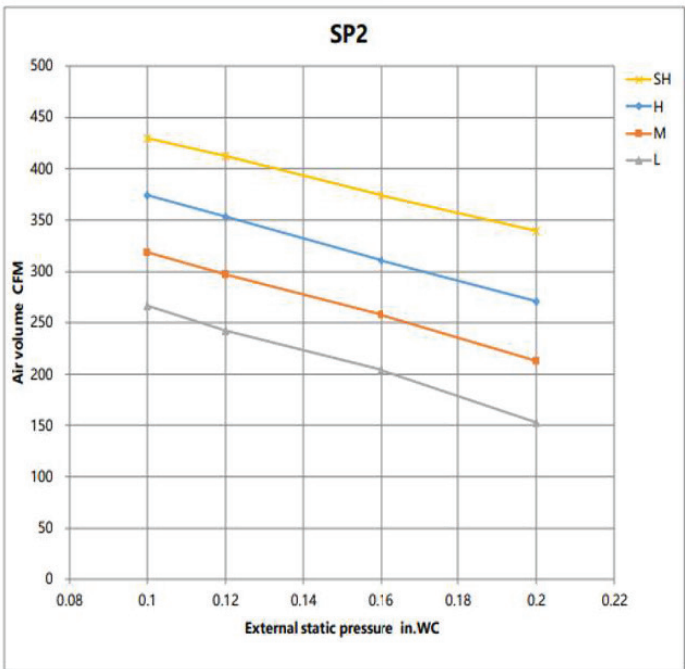
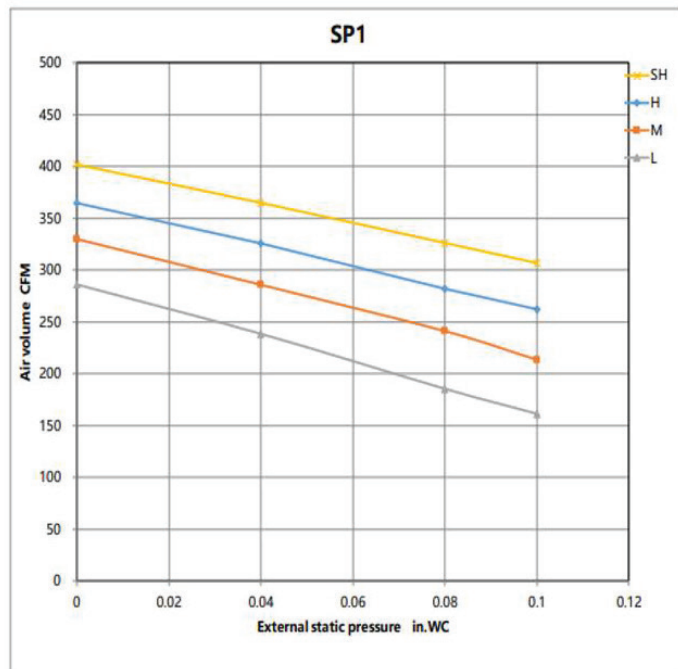


Outdoor Unit Dimensions and Connections

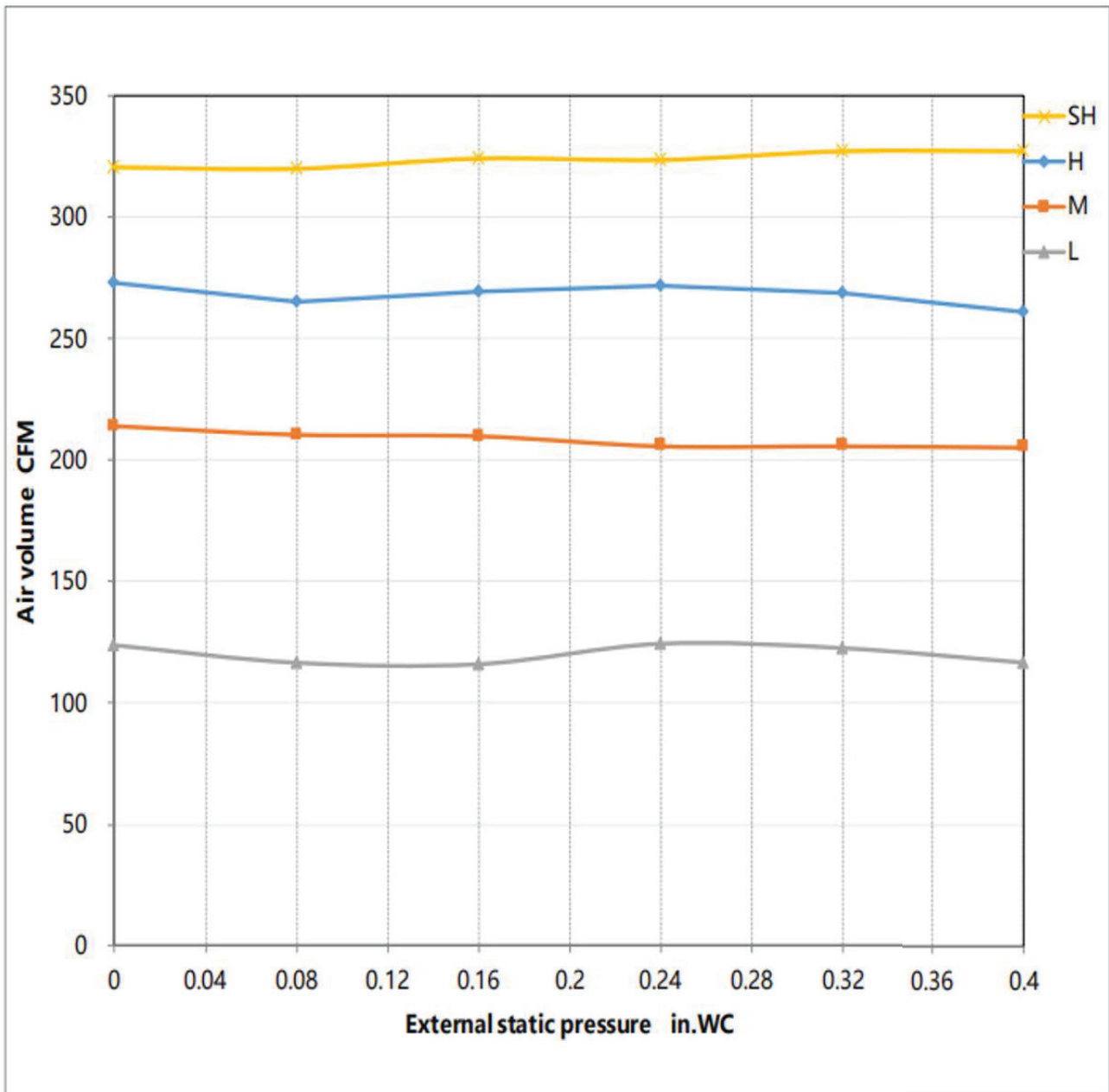


Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS009-1CSXHD	31.69	12.99	21.81	34.4	32.1	2.4	3.8	20.1	12.5	13.6

Air Volume



Air Volume



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Cooling Performance - Indoor Dry Bulb Temperature at 324 CFM																					
ODU DB(°F)	IDU WB (°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB (°F)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)	73.4°F (23.0°C)	77.0°F (25.0°C)	80.0°F (26.7°C)	80.6°F (27.0°C)	84.2°F (29.0°C)
-22°F (-30.0°C)	TC	9916	9911	10053	10095	10372	10346	10337	10331	10330	10321	10655	10642	10632	10630	10620	11239	11236	11213	11209	11185
	S/T	0.80	0.91	0.99	0.99	0.64	0.75	0.84	0.86	0.97	0.55	0.65	0.74	0.75	0.86	0.39	0.47	0.55	0.57	0.67	
	PI	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	
-13°F (-25.0°C)	TC	10742	10737	10891	10936	11237	11208	11199	11192	11191	11181	11543	11529	11518	11516	11505	12176	12173	12148	12143	12117
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	
-4°F (-20.0°C)	TC	11569	11563	11728	11777	12101	12070	12060	12053	12052	12042	12431	12416	12404	12402	12390	13112	13109	13082	13077	13049
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.62	0.62	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.62	0.62	0.62	0.62	0.62
0°F (-17.8°C)	TC	11953	11946	12167	12218	12554	12465	12451	12442	12440	12566	12832	12813	12798	12796	12790	13522	13519	13517	13519	13528
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.63	0.63	0.63	0.63	0.63
5°F (-15.0°C)	TC	12295	12338	12620	12678	13069	12872	12875	12876	12889	13081	13284	13287	13290	13290	13308	14063	14058	14066	14067	14074
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.65	0.65	0.65	0.65	0.65
14°F (-10.0°C)	TC	11188	11182	11410	11464	11819	11743	11733	11725	11723	11833	12145	12130	12118	12116	12104	12914	12910	12886	12881	12854
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56
17°F (-8.3°C)	TC	10794	10829	11079	11133	11495	11331	11336	11338	11348	11507	11718	11722	11727	11728	11742	12462	12459	12457	12459	12470
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.54	0.54	0.54	0.54	0.54
23°F (-5.0°C)	TC	11005	11052	11338	11394	11770	11562	11566	11567	11582	11782	11961	11965	11968	11968	11988	12718	12714	12723	12725	12732
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.55	0.55	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.57	0.57	0.57	0.57	0.57
35°F (1.7°C)	TC	11534	11534	11602	11634	12009	12073	12072	12073	12073	12105	12451	12450	12450	12451	12452	13152	13149	13148	13147	13148
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.56	0.56	0.57	0.57	0.59	0.60	0.60	0.60	0.60	0.60	0.62	0.62	0.62	0.62	0.62	0.68	0.68	0.68	0.68	0.68
41°F (5.0°C)	TC	11672	11672	11765	11846	12197	12214	12214	12215	12258	12596	12595	12595	12595	12596	12596	13302	13299	13298	13298	13299
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.55	0.55	0.56	0.56	0.58	0.59	0.59	0.59	0.59	0.59	0.61	0.61	0.61	0.61	0.61	0.67	0.67	0.67	0.67	0.67
47°F (8.3°C)	TC	11809	11817	11976	12029	12385	12357	12356	12357	12357	12414	12741	12740	12740	12740	12740	13454	13451	13449	13449	13450
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.54	0.54	0.55	0.55	0.58	0.58	0.58	0.58	0.58	0.58	0.60	0.60	0.60	0.60	0.60	0.66	0.66	0.66	0.66	0.66
50°F (10.0°C)	TC	11381	11694	12089	12159	12631	12024	12030	12113	12152	12643	12493	12509	12516	12516	12637	13354	13389	13424	13428	13444
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59	0.59
59°F (15.0°C)	TC	10733	11110	11504	11574	12045	11357	11372	11501	11587	12057	11825	11837	11841	11858	12067	12698	12723	12748	12751	12762
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.55	0.55	0.55	0.55	0.56	0.55	0.55	0.55	0.55	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56
68°F (20.0°C)	TC	10085	10515	10905	10975	11440	10677	10701	10917	10987	11452	11138	11146	11166	11176	11461	12001	12024	12042	12044	12051
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.51	0.51	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.53	0.53	0.53	0.53	0.53
77°F (25.0°C)	TC	9433	9888	10275	10343	10805	10001	10032	10286	10355	10816	10454	10465	10492	10504	10825	11290	11324	11351	11353	11364
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.49	0.49	0.49	0.49	0.49
82°F (27.8°C)	TC	9591	10081	10488	10559	11042	10164	10203	10497	10571	11054	10637	10650	10683	10696	11063	11500	11544	11574	11577	11589
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.56	0.57	0.57	0.57	0.57	0.57
86°F (30.0°C)	TC	10407	10926	11373	11452	11986	10975	11025	11386	11466	11998	11495	11508	11550	11567	12009	12444	12493	12522	12525	12537
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	0.74	0.74	0.75	0.75	0.75	0.74	0.74	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0.76	0.76	0.76	0.76	0.76
95°F (35.0°C)	TC	12171	12237	12632	12707	13209	12935	12943	12947	129629											

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Heating Performance - Indoor Dry Bulb Temperature at 324 CFM										
Outdoor DB(°F)	TCI Total Heating Capacity(BTU/hr)					PI Power Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-22°F(-30.0°C)	6582	6450	6418	6392	6335	1.46	1.47	1.47	1.48	1.48
-13°F(-25.0°C)	7863	7705	7667	7636	7567	1.57	1.58	1.58	1.59	1.59
-4°F(-20.0°C)	9952	9752	9704	9665	9578	1.68	1.69	1.69	1.70	1.70
0°F(-17.8°C)	10630	10417	10365	10323	10231	1.70	1.71	1.72	1.72	1.73
5°F(-15.0°C)	11477	11248	11191	11147	11046	1.73	1.75	1.75	1.75	1.76
14°F(-10.0°C)	12650	12397	12335	12286	12175	1.79	1.81	1.81	1.82	1.82
17°F(-8.3°C)	13041	12780	12717	12666	12552	1.81	1.83	1.83	1.84	1.84
22°F(-5.6°C)	13467	13198	13132	13079	12961	1.78	1.79	1.80	1.80	1.81
27°F(-2.8°C)	13893	13615	13547	13493	13371	1.75	1.76	1.76	1.77	1.77
32°F(0.0°C)	14318	14032	13962	13906	13781	1.71	1.73	1.73	1.73	1.74
35°F(1.7°C)	14574	14282	14211	14154	14027	1.69	1.70	1.71	1.71	1.72
37°F(2.8°C)	14909	14610	14537	14479	14349	1.64	1.66	1.66	1.66	1.67
42°F(5.6°C)	15745	15431	15353	15292	15154	1.53	1.54	1.54	1.55	1.55
44.6°F(7.0°C)	16080	15759	15680	15617	15477	1.48	1.50	1.50	1.50	1.51
47°F(8.3°C)	16582	16251	16169	16105	15960	1.41	1.43	1.43	1.43	1.44
52°F(11.1°C)	15876	15558	15480	15418	15280	1.26	1.28	1.28	1.28	1.29
57°F(13.9°C)	15169	14865	14791	14732	14599	1.12	1.13	1.13	1.13	1.13
59°F(15.0°C)	14886	14588	14515	14457	14327	1.06	1.07	1.07	1.07	1.07
62°F(16.7°C)	14462	14173	14102	14045	13919	0.97	0.98	0.98	0.98	0.98
64.4°F(18.0°C)	14122	13840	13771	13716	13592	0.90	0.90	0.91	0.91	0.91
69.8°F(21.0°C)	12292	12047	11986	11938	11831	0.74	0.74	0.74	0.74	0.75
75.2°F(24.0°C)	9154	8971	8926	8890	8810	0.57	0.58	0.58	0.58	0.58

Bolded values were tested. All other values are simulated and for reference only.